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→ Regression

From bivariate data given by (x, y) the estimation/prediction of avg value of variable say y corresponding to other variable x is called regression.

This is done by a suitable curve/straight line which fits best on scatter diagram of data. This line is called regression line / least sq regression line

Regression line which is fit to find avg value of y depending on x is called as regression line of y on x. equation of this line is called as regression equation

→ equation of regression line

Let $(x_1 y_1)$ $(x_2 y_2)$ . . . . $(x_n y_n)$ be n pairs of observations given by bivariate data (x, y)

Let $\bar{x}, \bar{y}$ be means & σx σy be standard deviations of x & y respectively, r_{xy} is co-relation coefficient then

1. Regression line of y on x

$$y - \bar{y} = \underbrace{b_{yx}}_{\text{regression coeff of } y \text{ on } x} (x - \bar{x}) \qquad b_{yx} = r_{xy} \frac{\sigma y}{\sigma x}$$

2. Regression line of x on y

$$x - \bar{x} = \underbrace{b_{xy}}_{\text{regression coeff of } x \text{ on } y} (y - \bar{y}) \qquad b_{xy} = r_{xy} \frac{\sigma x}{\sigma y}$$

→ Properties of regression line & coefficients

- b_{xy} & r_{xy} have same sign
- $b_{xy} \times b_{yx} = r^2_{xy}$
- The 2 regression lines intersect at point $(\bar{x}, \bar{y})$
- Gradient of regression line of y on x is b_{xy} and of x on y is $\frac{1}{b_{xy}}$

Q Find two regression equations. Hence estimate purchase when sale will be 100. To get purchase of 60 what is required sales?

S Sales	P Purchase	$u = x - 90$	$v = y - 70$	u^2	v^2	uv
91	71	1	1	1	1	1
97	75	7	5	49	25	35
108	69	18	-1	324	1	-18
121	97	31	27	961	729	837
67	70	-23	0	529	0	0
124	91	34	21	1156	441	714
51	39	-39	-31	1521	961	1209
73	61	-17	-9	289	81	153
111	80	21	10	441	100	210
57	47	-33	-23	1089	529	759
		0	0	6360	2868	3900

$\bar{u} = 0 \quad \bar{v} = 0 \quad \bar{x} = 90 \quad \bar{y} = 70$

$var(u) = \frac{\Sigma u^2}{10} = 636 \qquad Var(y) = \frac{\Sigma v^2}{10} = 286.8$

$\sigma(x) = \sigma(u) = \sqrt{636} \qquad \sigma(y) = \sigma(v) = \sqrt{286.8}$

$\sigma x = 25.21 \qquad \sigma y = 16.93$

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Now,

$$cov(u,v) = \frac{\Sigma u_i v_i}{10} - 0$$

$$= 390$$

$$\Rightarrow r_{uv} = \frac{390}{(25.21)(16.93)} = 0.91$$

$$\Rightarrow r_{xy} = + r_{uv} = + 0.91$$

$$\Rightarrow b_{xy} = \frac{\sigma x}{\sigma y} r_{xy}$$

$$b_{xy} = (1.48)(0.91) = 1.35$$

$$\& \; b_{yx} = \frac{\sigma y}{\sigma x} r_{xy}$$

$$b_{yx} = (0.67)(0.91) = 0.61$$

$$\Rightarrow y - \bar{y} = b_{yx}(x - \bar{x})$$

$$\Rightarrow y - 70 = 0.61(x - 90)$$

and

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

$$x - 90 = 1.35(y - 70)$$

$\Rightarrow$ when $x = 100$

$y - 70 = 10 \div 1.35 = 7.40$

$\Rightarrow y = 77.40$

$\Rightarrow$ when $y = 60$

$x - 90 = -10 \div 0.61 = -16.39$

$x = 73.60$

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Q $x + 4y + 3 = 0$ and $4x + 9y + 5 = 0$ are two regression lines. Find which one is of y on x. Find mean of x & y; r_{xy}, value of x when $y = 1.5$

$\hookrightarrow x = -9$

Let's assume $x + 4y + 3 = 0$ is line for y on x

$$y = -\frac{1}{4}x - \frac{3}{4}$$

$m = -1/4$ ↳ slope

$$\Rightarrow b_{xy} = -\frac{1}{4}$$

$$\frac{\sigma_x}{\sigma_y} r_{xy} = -\frac{1}{4} \quad -①$$

$$4x + 9y + 5 = 0$$

$$y = -\frac{4x}{9} - \frac{5}{9}$$

$$m = -4/9$$

$$\Rightarrow 1/b_{yx} = -4/9$$

$$\Rightarrow \frac{\sigma_y}{\sigma_x} r_{xy} = -\frac{9}{4} \quad -②$$

① × ②

$$\Rightarrow r^2_{xy} = -\frac{1}{4} \times \left(-\frac{9}{4}\right) = \frac{9}{16}$$

$$\Rightarrow r_{xy} = \pm\frac{3}{4} = \pm 0.75 \quad \Rightarrow r_{xy} = -0.75$$

as $b_{xy} \cdot b_{yx} = r^2_{xy}$ and all b_{yx}, b_{xy} and r_{xy} has same sign and $-1 \le r_{xy} \le 1$ Hence our assumption is right (all properties satisfied)

$\Rightarrow r_{xy} = -0.75$

$\Rightarrow x + 4y + 3 = 0$ is reg. line for y on x

↳ ×4

$$4x + 16y = -12$$
$$4x + 9y = -5$$
$$7y = -7$$
$$y = -1 \Rightarrow x = 1$$

pt of intersection $= (\bar{x}, \bar{y}) = (1, -1)$

$\Rightarrow \bar{x} = 1$ & $\bar{y} = -1$

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Q. If θ is acute angle between 2 regression lines show that

$$\tan\theta = \frac{1-r^2}{r} \cdot \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2}$$

slope for y on x = b_{yx}

$$m_1 = \frac{\sigma_y}{\sigma_x} r_{xy}$$

slope for x on y = b_{xy}

$$m_2 = 1 \Big/ \frac{\sigma_x}{\sigma_y} r_{xy} = \frac{\sigma_y}{r_{xy}\sigma_x}$$

Now $\tan\theta = \frac{|m_1 - m_2|}{1 + m_1 m_2}$

$$= \frac{\left(\frac{\sigma_y r}{\sigma_x} - \frac{\sigma_y}{r\sigma_x}\right)}{1 + \frac{\sigma_y^2}{\sigma_x^2}}$$

$$= \frac{\frac{\sigma_y}{\cancel{\sigma_x}}\left(\frac{r^2-1}{r}\right)}{\frac{\sigma^2_x + \sigma^2_y}{\sigma^{\cancel{2}}_x}}$$

$$= \frac{(r^2-1)\,\sigma_y\,(\sigma_x)}{(\sigma^2_x + \sigma^2_y)\,r}$$

$$= \frac{|r^2-1|}{r} \quad \frac{\sigma_y \sigma_x}{(\sigma^2_x + \sigma^2_y)}$$

$\longrightarrow$ Hence Proved

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☆☆

→ Curve Fitting

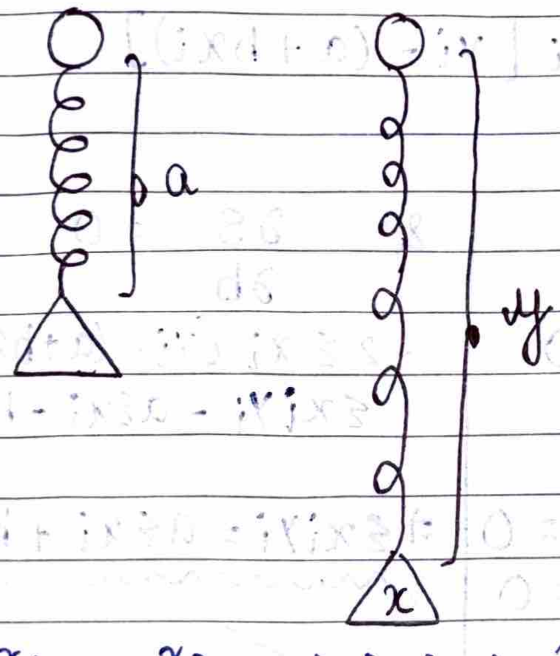
We find a curve called best fitted curve $y = \phi(x)$, we can predict value of y to corresponding value of x

$\underbrace{y = \phi(x)}_{\text{Best fitted curve of } y \text{ on } x}$ and $\underbrace{y = \psi(x)}_{\text{Best fitted curve of } x \text{ on } y}$

→ Method of least square

Distance between all the points in scatter diagram is minimum from regression line

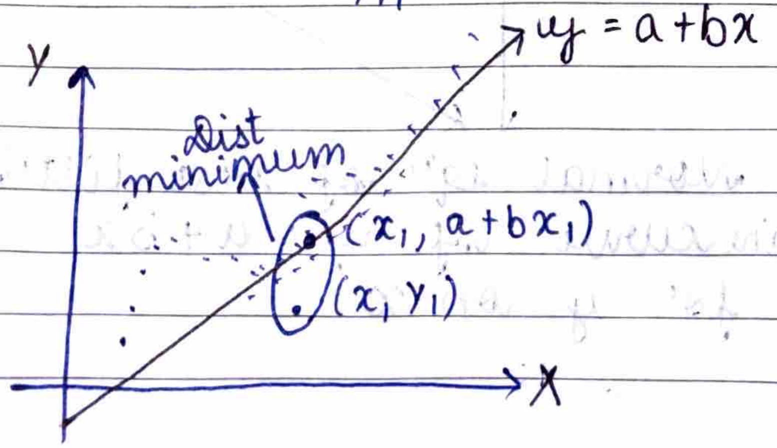
eg of a spring



$a \rightarrow$ normal length
$y \rightarrow$ increased length
$x \rightarrow$ weight added

$y - a \propto x$
$y - a = bx$
$y = a + bx$

x_1	x_2	. . . . .	x_n
y_2	y_2		y_n



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Hence, sum of difference of heights for all points

$$S(a,b) = \underbrace{\sum_{i=1}^{n} [y_i - (a+bx_i)]^2}_{\downarrow}$$

→ downward dist -ve aur upward +ve ati sabko same sign ke lie SBS kardia

we need to minimise this

Diff partially wrt a & b

$$\Rightarrow \frac{\partial S}{\partial a} = -2 \sum_{i=1}^{n} [y_i - (a+bx_i)]$$

$$\Rightarrow \frac{\partial S}{\partial b} = -2x_i \sum_{i=1}^{n} [y_i - (a+bx_i)]$$

$$= -2 \sum_{i=1}^{n} x_i [y_i - (a+bx_i)]$$

NOW $\frac{\partial S}{\partial a} = 0$

$\Rightarrow -2 \sum y_i - (a+bx_i) = 0$

$\Rightarrow \sum y_i - (a+bx_i) = 0$

$\sum y_i - \sum a - \sum bx_i = 0$

$\sum y_i - na - b\sum x_i = 0$

$\sum y_i = na + b\sum x_i$

& $\frac{\partial S}{\partial b} = 0$

$-2 \sum x_i [y_i - (a+bx_i) = 0$

$\sum x_i y_i - a\sum x_i - b\sum x_i^2 = 0$

$\Rightarrow \sum x_i y_i = a\sum x_i + b\sum x_i^2$

Normal eq^n of best fitting st line curve of $y = a + bx$ for y on x

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$na + b\Sigma x_i - \Sigma y_i = 0$

$a\Sigma x_i + b\Sigma x_i^2 - \Sigma x_i y_i = 0$

using cross multiplication for 2 variables a and b

$$\frac{a}{-\Sigma x_i \Sigma x_i y_i + \Sigma y_i \Sigma x_i^2} = \frac{b}{-n\Sigma x_i y_i + \Sigma x_i y_i} = \frac{1}{n\Sigma x_i^2 - (\Sigma x_i)^2}$$

$$\Rightarrow b = \frac{\Sigma x_i y_i - n\Sigma x_i y_i}{n\Sigma x_i^2 - (\Sigma x_i)^2} \quad \times, \div \text{ by } \frac{1}{n^2}$$

$$\Rightarrow b = \frac{\left(\frac{1}{n}\Sigma x_i \cdot \frac{1}{n}\Sigma y_i\right) - \frac{1}{n}\Sigma x_i y_i}{\frac{1}{n}\Sigma x_i^2 - \left(\frac{1}{n}\Sigma x_i\right)^2}$$

$$= \frac{\frac{1}{n}\Sigma x_i y_i - \bar{x}\bar{y}}{\frac{1}{n}\Sigma x_i^2 - \left(\frac{1}{n}\Sigma x_i\right)^2} = \frac{cov(x,y)}{var(x)}$$

$$\Rightarrow b = \frac{cov(x,y)}{var(x)}$$

Now, $\Sigma y_i = na + b\Sigma x_i$ $\quad \div$ BS by n

$\bar{y} = a + b\bar{x}$

$a = \bar{y} - b\bar{x}$

Put values in $y = a + bx$

$$y = \bar{y} - b\bar{x} + bx$$

$$y - \bar{y} = b(x - \bar{x})$$

$$\Rightarrow y - \bar{y} = \frac{cov(x, y)}{var(x)} (x - \bar{x})$$

Regression line / least square line of y on x

or Best fitted line of $y = a + bx$

// ly

$$\left.\begin{aligned} x &= a + by \\ \Sigma x_i &= na + b\Sigma y_i \\ \Sigma x_i y_i &= a\Sigma y_i + b\Sigma y_i^2 \end{aligned}\right\}$$ normal eqn for regression line of x on y

Q Fit a straight line $y = a + bx$ to data

X	2	5	6	8	9
Y	8	14	19	20	31

Predict y when $x = 9.6$

X	Y	$x_i y_i$	x_i^2
2	8	16	4
5	14	70	25
6	19	114	36
8	20	160	64
9	31	279	81
$\Sigma x_i = 30$	$\Sigma y_i = 92$	$\Sigma x_i y_i = 639$	$\Sigma x_i^2 = 2$[illegible]0

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Normal eq

$$\left.\begin{aligned}\Sigma y_i &= na + b\Sigma x_i \\ \Sigma x_i y_i &= a\Sigma x_i + b\Sigma x_i^2\end{aligned}\right\} \quad n = 5$$

$\Rightarrow$ $92 = 5a + 30b$ & $639 = 30a + 210b$

$184 = 10a + 60b$ $\quad$ $213 = 10a + 70b$

$213 = 10a + 70b$

$\underline{184 = 10a + 60b}$

$29 = 10b$ $\quad$ $10a = 213 - 203 = 10$

$b = 2.9$ $\quad$ $a = 1$

$\Rightarrow$ eq^n of best fitted st line of form

$y = a + bx$ is

$y = 1 + 2.9x$

when $x = 9.6$

$y = 1 + 2.9(9.6)$

$= 28.84$

$\rightarrow$ Finding best fitted parabola

Best fitted second degree parabola will be

$y = a + bx + cx^2$

$$\left.\begin{aligned}an + b\Sigma x_i + c\Sigma x_i^2 &= \Sigma y_i \\ a\Sigma x_i + b\Sigma x_i^2 + c\Sigma x_i^3 &= \Sigma x_i y_i \\ a\Sigma x_i^2 + b\Sigma x_i^3 + c\Sigma x_i^4 &= \Sigma x_i^2 y_i\end{aligned}\right\}$$

Normal equations

for $y = a + bx + cx^2$

Q Find $y = a + bx + cx^2$ $x = 4.8$ $y = ?$

x	y	x^2	x^3	x^4	xy	x^2y
0	1	0	0	0	0	0
1	5	1	1	1	5	5
2	10	4	8	16	20	40
3	22	9	27	81	66	198
4	38	16	64	256	152	608
10	76	30	100	354	243	851

$\Sigma y_i = na + b\Sigma x_i + c\Sigma x_i^2$

$76 = 5a + 10b + 30c$ - ①

$\Sigma x_i y_i = a\Sigma x_i + b\Sigma x_i^2 + c\Sigma x_i^3$

$243 = 10a + 30b + 100c$ - ②

$\Sigma x_i^2 y_i = a\Sigma x_i^2 + b\Sigma x_i^3 + c\Sigma x_i^4$

$851 = 30a + 100b + 354c$ - ③

$729 = 30a + 90b + 300c$

$122 = 10b + 54c$

$61 = 5b + 27c$

$5b = 61 - 27c$ - ④

$243 = 10a + (366 - 162c) + 100c$

$-123 = 10a - 62c$

$10a = 62c - 123$ - ⑤

$152 = 10a + 20b + 60c$

$152 = 62c - 123 + 244 - 108c + 60c$

$152 = 14c + 121$

$31 = 14c$

$c = 2.21$

$\Rightarrow b = 0.26$

$\Rightarrow a = 1.40$

$y = 1.40 + 0.26x + 2.21x^2$

$x = 4.8,\ y = 53.56$

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Q Fit an exponential curve of form $y = ab^x$ to following data

x	y	$\log_{10} y = Y$	$x_i Y$	x_i^2
2	640	2.80	5.61	4
3	512	2.70	8.12	9
4	410	2.61	10.45	16
5	328	2.51	12.57	25
6	262	2.41	14.50	36
7	210	2.32	16.25	49
27		15.38	67.53	139

$$y = ab^x$$

Take $\log_{10}$

$$\log_{10} y = \log_{10} a + x \log_{10} b$$

$$Y = A + Bx$$

It's linear now

$$\Rightarrow \Sigma Y_i = nA + B\Sigma x_i$$

$$\& \ \Sigma x_i Y_i = A\Sigma x_i + B\Sigma x_i^2$$

$\Rightarrow$ $15.38 = 6A + 27B$ $\quad \times 9$ $\qquad$ $67.53 = 27A + 139B$ $\quad \times 2$

$$138.42 = 54A + 243B$$

$$135.06 = 54A + 278B$$

$$3.36 = -35B$$

$$\Rightarrow B = -0.096$$

$\Rightarrow$ $A = 2.99$

$2.99 = \log_{10} a$ $\qquad$ $-0.096 = \log_{10} b$

$a = (10)^{2.99}$ $\qquad$ $b = (10)^{-0.096}$

$a = 977.23$ $\qquad$ $b = 0.801$

$$\Rightarrow y = (977.23)(0.801)^x$$

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Q Fit a geometric curve of form $y = ax^b$

x	y	$X = \log_{10} x$	$Y = \log_{10} y$	X^2	XY
1	5.0	0	0.69	0	0
2	6.3	0.30	0.79	0.09	0.23
3	7.2	0.47	0.85	0.22	0.39
4	7.9	0.60	0.89	0.36	0.53
5	8.5	0.69	0.92	0.47	0.82
		2.06	4.14	1.14	1.78

$$\log_{10} y = \log_{10} a + b \log_{10} x$$

$$\Rightarrow Y = A + bX$$

$$\Rightarrow \Sigma Y = nA + b\Sigma X \quad \& \quad \Sigma XY = A\Sigma X_i + b\Sigma X^2$$

$$\Rightarrow 4.14 = 5A + 2.06b \quad \& \quad 1.78 = 2.06A + 1.14b$$

$$\times 2.06 \qquad\qquad \times 5$$

$$\Rightarrow 9.08 = \cancel{10.3A} + 4.24b$$

$$8.92 = \cancel{10.3A} + 5.7b$$

$$= -1.46b = 0.156$$

$$\Rightarrow b = -0.106$$

$$\Rightarrow A = 0.922$$

$$\log_{10} a = 0.922$$

$$a = (10)^{0.922}$$

$$a = 8.370$$

$$\Rightarrow y = (8.37)(x)^{-0.106}$$

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→ Test of significance

Large sample Test
N > 30 (Z test)

- Single Mean
- Difference of Mean
- Single Proportion
- Difference of proportion
- Single standard deviation (Chi square test)
- Difference of standard deviation

Small sample Test
N ≤ 30 (t test)

- Single Mean
- Difference of means
- Chi square test for goodness of fit

- Each test can be done using 5 steps
1. Gather all the infromation from ques
2. Write hypothesis & identify one tailed or two tailed test (for Z test) only)
3. Write critical values
4. Compare all kinds of values
   - If test statistic < critical value, that means H0 is accepted and H1 is rejected
   - If test statistic > critical value, that means H0 is rejected & H1 is accepted
5. Final verdict

Z - test (large samples)

one tailed test / one sided alternative hypothesis (> or <)

- 1% level of significance — 2.33
- 5% level of significance — 1.64

two sided alternative hypothesis / 2 tailed test (≠)

- 1% level of significance — 2.576
- 5% level of significance — 1.96

Learn →

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Hypothesis → Write always about population parameters (μ, σ etc)

Null Hypothesis (H_0)

Alternative Hypothesis (H_1)

used for comparison
eg → to test less than, more than, not equals to, not equal to

Critical Value

Z test ↓ If not given use z test flowchart (prev page)

t / χ^2 test ↓ given in ques

→ Large sample test for single mean

Q A machine part was designed to withstand an avg pressure of 120 units. A random sample of size 100 from large batch was tested and it was found that avg pressure these parts can withstand is 105 units with standard deviation of 20 units. Test at 5% level whether batch meet specification? Suppose population has normal distribution

Population mean = μ = 120

$N = 100$ = sample size

Sample mean = 105 = $\bar{x}$

sample standard deviation = 20

For large sample σ (population) $\cong$ σ (sample)

$\Rightarrow$ σ pop = 20 & level of significance = 5%

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→ Null Hypothesis
$H_0 : \mu = 120$
Batch meets specification
→ Alternative Hypothesis
$H_1 : \mu < 120$ → one tailed test
Batch doesn't meet specification

Now critical value isn't given, using flowchart
$Z_{crit} = 1.64$

Test statistic $= \left| \dfrac{\bar{x} - \mu}{\sigma / \sqrt{n}} \right| = Z_{cal}$

⇒ Test stat $= \left| \dfrac{105 - 120}{20 / \sqrt{100}} \right| = 7.5$

$7.5 > 1.64$
$Z_{cal} > Z_{crit}$

H_0 is rejected & H_1 is accepted
Therefore, the batch doesn't meet specification

→ Large sample test for difference of means
Q LED bulbs manufactured by companies A & B gave following results

	A	B
No of bulbs used	100	100
Avg life (in hour)	1248	1300
s.d (in hour)	93	82

Find whether avg life of 2 bulbs made by two comapanies A & B are same. Test at 5% level of significance given area under normal curve

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enclosed between $z = 1.96$ and $z = \infty$ is 0.025

A	B
$n_1 = 100$	$n_2 = 100$
$\bar{x}_1 = 1248$	$\bar{x}_2 = 1300$
$\sigma_1 = 93$	$\sigma_2 = 82$

Null Hypothesis

$H_0 : \mu_1 = \mu_2 \Rightarrow \mu_1 - \mu_2 = 0$

Yes they are same

Alternative Hypothesis

$H_1 : \mu_1 \neq \mu_2$ → Two tailed test

No they aren't same

Now, $z_{crit} = 1.96$

$\mu_1 - \mu_2 = 0$ (always)

$$\text{test stat} = \left| \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \right|$$

$$= \left| \frac{-52 - 0}{\sqrt{\frac{93^2 + 82^2}{100}}} \right| \cong 4.19$$

$z_{cal} = 4.19$

$4.19 > 1.96$

$\Rightarrow H_1$ is accepted

Therefore avg life of 2 companies are not same.

→ Large Sample test for single proportion

Q A sample of size 600 persons selected at random from a large city shows that % of males in sample is 53%. It's believed that male to

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total proportion ratio in city is 1/2. Test whether this belief is confirmed by observation. Given that $\left[\int_{-\infty}^{-1.96} \phi(z)\, dz = 0.025\right]$

$n = 600$

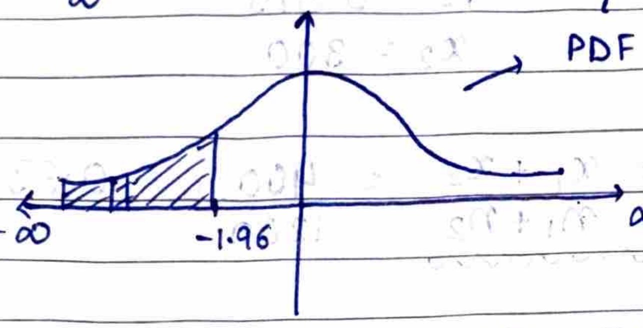
sample proportion of male $= \hat{P} = 0.53$

Population proportion of male $= P = 0.5$

Now, $\int_{-\infty}^{-1.96} \phi(z)\, d(z) = 0.025$

Probablity dist func

PDF of standard normal distribution



$\frac{\alpha}{2} = 0.025 \Rightarrow \alpha = 0.05 = 5\%$ level of sig

level of significance

Null Hypothesis

$H_0 : P = \hat{P}$ Belief is right

Alternative Hypothesis

$H_1 : P \neq \hat{P}$ Belief isn't right $\longrightarrow$ 2 tailed test

$\Rightarrow Z_{crit} = 1.96$

Now

test statistic $= \left| \dfrac{\hat{P} - P}{\sqrt{\dfrac{P(1-P)}{n}}} \right|$

$Z_{cal} \approx 1.47$

$1.47 < 1.96 \Rightarrow H_0$ is accepted

Therefore belief is right

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→ Large sample test for difference of proportions

Q In a certain city 100 men in sample of 400 were found to be smokers. In another city no of smokers were 300 in sample of 800. Does this indicate that there is a greater proportion of smokers in second city than in first city? Test at 1% level of significance

$n_1 = 400$ $\quad\quad$ $n_2 = 800$

$\hat{P}_1 = 0.250$ $\quad\quad$ $\hat{P}_2 = 0.375$

$x_1 = 100$ $\quad\quad$ $x_2 = 300$

Now $P = \frac{x_1 + x_2}{n_1 + n_2} = \frac{400}{1200} = 0.33$

Null hypothesis

$H_0 : \hat{P}_1 = \hat{P}_2$ Proportion of smokers in second city isn't greater

$H_1 : \hat{P}_2 > \hat{P}_1$, Proportion of smokers in second city is greater

one tailed test

$Z_{crit} = 2.33$

Z_{cal} = test stat $= \left| \frac{\hat{P}_1 - \hat{P}_2}{\sqrt{P(1-P)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \right|$

$= \left| \frac{0.25 - 0.375}{\sqrt{0.33(0.66)\left(\frac{3}{800}\right)}} \right| = 9.1856$

$Z_{cal} > Z_{crit}$

H_1 is accepted. Therefore proportion of smoker in second city is greater than first city

χ^2 test 'chi-square'

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Large sample test for single standard deviation

Q. A random sample of size 31 from a normal population gives a sample mean 42 and sample standard dev of 5. Test at 5% level of significance that population sd is 7. given $\chi^2_{0.05} = 43.77$ with 30 degrees of freedom.

$n = 31$ $\quad\quad \sigma_p = 7$

$\bar{x} = 42$ $\quad\quad$ degree of freedom $= n-1$

$\sigma_s = 5$

Null Hypothesis

$H_0 : \sigma_p = 7$ population sd maybe 7

Alternative Hypothesis

$H_1 : \sigma_p \neq 7$ population sd mayn't be 7

Now, $\chi^2_{0.05} = \chi^2 crit = 43.77$

Test stat $= \left| \frac{\Sigma(x_i - \bar{x})^2}{\sigma_p^2} \right|$

Also $\sigma_s = \frac{\Sigma x_i - \bar{x}}{n}$

$\Rightarrow n^2 \sigma_s^2 = \Sigma(x_i - \bar{x})^2$

$\Rightarrow$ Test stat $= \left| \frac{n^2 \sigma_s^2}{\sigma_p^2} \right|$

$\Rightarrow \chi^2 cal = \left| \frac{31^2 \times 5^2}{7^2} \right| = 15.82$

$\chi^2 cal < \chi^2 crit$

$\Rightarrow$ Ho is accepted

Therefore population sd maybe 7

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→ Large sample test for difference of sd.

Q Intelligence test on 2 groups of boys & girls gave following results

	Mean	SD	Number
Boys	70	20	250
Girls	75	15	150

Is there a significant difference in the standard deviation of scores obtained by boys and girls. Test at 1% los.

$n_1 = 250$ $\quad$ $n_2 = 150$

$\bar{x}_1 = 70$ $\quad$ $\bar{x}_2 = 75$

$\sigma_1 = 20$ $\quad$ $\sigma_2 = 15$

Null Hypothesis

$H_0 : \sigma_1 = \sigma_2$ $\quad$ No significant difference

Alternative Hypothesis

$H_1 : \sigma_1 \neq \sigma_2$ $\quad$ yes significant difference

2 tailed test

$Z_{crit} = 2.576$

$$\text{test stat} = \left| \frac{\sigma_1 - \sigma_2}{\sqrt{\frac{\sigma_1^2}{2n_1} + \frac{\sigma_2^2}{2n_2}}} \right|$$

$Z_{cal} = 4.016$

$Z_{cal} > Z_{crit}$

H_1 is accepted

Therefore, there is significant difference in sd of scores of boys & girls.

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↱ t - test

→ small sample test for single mean

• case I (σ population not given) → use t test

Q certain pesticide is packed into bags by machine. A random sample of 10 bags is drawn & their contains are found to weigh 50, 49, 52, 44, 45, 48, 46, 45, 49, 45. Test if average packing can be taken to be 50 Kg. $t_{0.25,9} = 2.26$

$$t = \left| \frac{\bar{x} - \mu}{\sigma_s / \sqrt{n-1}} \right|$$

$$\bar{x} = \frac{50+49+52+44+45+48+46+45+49+45}{10}$$

$\bar{x} = 47.3$ kgs

x_i	$d = x_i - 47$	d^2
50	3	9
49	+2	4
52	+5	25
44	-3	9
45	-2	4
48	+1	1
46	-1	1
45	-2	4
49	2	4
45	-2	4
	3	67

$\bar{x} = 47 + \frac{3}{10} = 47.3$ $\bar{Y} = 0.3$

$$\sigma_y = \sqrt{\frac{\Sigma y_i^2}{n} - \bar{y}^2} = \sqrt{6.7 - 0.09} = 2.57$$

$\Rightarrow \sigma_x = 2.57$

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$\Rightarrow t = \left| \dfrac{47.3 - 50}{2.57/\sqrt{9}} \right| = \left| \dfrac{-2.7 \times 3}{2.57} \right|$

$\Rightarrow t = 3.1517$

$t_{cal} = 3.15$

$t_{crit} = 2.26$

$t_{cal} > t_{crit}$

Null Hypothesis
$H_0 : \mu = 50$
Alternative
$H_1 : \mu \neq 50$

$t_{cal} > t_{crit}$
H_1 is accepted
$\Rightarrow$ Avg packing cant be 50 kgs

- Case II (σ population given) $\rightarrow$ use Z test

Q A tyre manufacturing company claims that average life of their product is 69000 miles. A car manufacturing company plans to purchase tyre, collects a sample of 10 tyre's life from 'population of life' of tyres which are 65, 71, 64, 71, 70, 69, 64, 63, 67 and 68 thousands of miles. It's known that population is normal & $\sigma_p = \sqrt{7.056}$. Test at 5% level of significance that avg life of tyre is lower than company claims. $Z_{0.5} = 1.65$, $Z_{0.1} = 2.36$
$t_{0.1,9} = 2.28$

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$\mu = 69000$ miles

$\sigma_p = \sqrt{7.056}$

$n = 10$

$$z = \left| \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \right|$$

(1000 miles)

x	$d = x - 66$
65	-1
71	5
64	-2
71	5
70	4
69	3
64	-2
63	-3
67	1
68	2
	12

$\bar{x} = \left(66 + \frac{12}{10}\right) 1000$ mil

$\bar{x} = (66 + 1.2) \times 1000$

$\bar{x} = 67.2 \times 1000$

$\Rightarrow \bar{x} = 67200$ miles

Zcrit = 1.65 thousand miles

Null

$H_0 : \mu = 69000$ miles

avg life of tyre of company is what it claim

Alternative

$H_1 : \mu < 69000$ miles $\leftrightarrow$ 1 tailed

avg life of tyre of company isn't what it claim

$$Zcal = \left| \frac{67200 - 69000}{\sqrt{0.7056}} \right| = \left| \frac{1800}{0.84} \right| = 2142.85$$

Zcal = 2.14 thousand miles

Zcal > Zcrit

H_1 is accepted

Avg life of tyre of company isn't what is claims.

→ small sample test for difference of means

• case I (σ_p is given) → use z test

Q There are 2 normal populations I and II having $\sigma_1 = 8$ and $\sigma_2 = 6$. Two independent samples of size 10 and 12 are drawn with sample mean 20 & 27. Test at 1% level

1) whether $\mu_1 = \mu_2$

11) $\mu_1 < \mu_2$

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$\sigma_1 = 8$ $\quad$ $\sigma_2 = 6$

$n_1 = 10$ $\quad$ $n_2 = 12$

$\bar{x}_1 = 20$ $\quad$ $\bar{x}_2 = 27$

1) Null

$H_0 : \mu_1 = \mu_2$

Alternative

$H_1 : \mu_1 \neq \mu_2$ → 2 tailed test

$z_{crit} = 2.576$

$$z_{cal} = \frac{-7}{\sqrt{6.4 + 3}} = \frac{-7}{\sqrt{9.4}} = -2.28$$

$|z_{cal}| < z_{crit}$

H_0 is accepted

Yes, $\mu_1 = \mu_2$

11) Null $H_0 : \mu_1 = \mu_2$

Alternat $H_1 : \mu_1 < \mu_2$ → 1 tailed test

$z_{crit} = 2.33$

$$z_{cal} = \frac{7}{\sqrt{9.4}} = -2.28$$

$z_{crit} > |z_{cal}|$ $\quad$ H_0 accepted.

No, μ_1 isn't less than μ_2

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- Case (II) [σ_P isn't given] → use t test

Q Two types of batteries are tested for their length of life distributed normally and following data are obtained

	A	B
No of Batteries	9	8
Mean life (hours)	600	640
Variance	121	144

Is there a significant difference in lifetimes of 2 type of batteries. Value of t for 15 degree of freedom at 5% level is 2.131

$$t = \sqrt{\frac{n_1 n_2 (n_1+n_2-2)}{n_1+n_2}} \cdot \left(\frac{\bar{x}_1 - \bar{x}_2}{\sqrt{n_1\sigma_{s_1}^2 + n_2\sigma_{s_2}^2}}\right)$$

$n_1 = 9$ $\quad$ $n_2 = 8$

$\bar{x}_1 = 600$ $\quad$ $\bar{x}_2 = 640$

$\sigma_{s_1} = 11$ $\quad$ $\sigma_{s_2} = 12$

$t_{crit} = 2.131$

Null Hypo

$H_0 : \mu_1 = \mu_2$ There isn't significant difference

$H_1 : \mu_1 \neq \mu_2$ There is significant difference

$$t_{cal} = \sqrt{\frac{72}{17}(15)}\left(\frac{-40}{\sqrt{9 \times 121 + 8 \times 144}}\right) = -6.73$$

$|t_{cal}| = 6.73$

$|t_{cal}| > t_{crit}$

H_1 is accepted

∴ There is significant difference in avg life time of batteries.

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→ χ^2 test for goodness of fit

$$\chi^2 = \frac{\Sigma (f_o - f_e)^2}{f_e} \rightarrow \text{Test statistic}$$

f_o → observed frequency
f_e → expected frequency

Q survey of 320 families with 5 children each revealed following distribution

No of Boys	5	4	3	2	1	0
No of girls	0	1	2	3	4	5
No of families	14	56	110	88	40	12

Is the result consistent with hypothesis that male & female births are equally probable? 5% value of χ^2 with 5 degree of freedom is 11·07

Here we think 'family with n boys' as the classes

⇒ No of classes = 6

Ho : Male & female are equally probable
P(male) = P(female)

out of 5 children

$P(0B) = \left(\frac{1}{2}\right)^5 = 1/32$ $\qquad P(1B) = \frac{1}{2} \times \left(\frac{1}{2}\right)^4 = \frac{1}{32}$

⇒ $P(nB) = \frac{1}{32} \, {}^5C_n$

∴ For 320 families, expected families having n boys

$= 320 \times \frac{1}{32} \times {}^5C_n = 10 \times {}^5C_n$

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No of Boys	fo	(P) Probablity	(P×320) fe	fo-fe	$(fo-fe)^2$	$\frac{(fo-fe)^2}{fe}$
0	12	1/32	10	-2	4	0.4
1	40	5/32	50	-10	100	2
2	88	10/32	100	-12	144	1.44
3	110	10/32	100	10	100	1
4	56	5/32	50	6	36	0.72
5	14	1/32	10	4	16	1.6
						7.16

$x^2 = 7.16$

$x^2_{cal} = 7.16$

$x^2_{crit} = 11.07$

$x^2_{crit} > x^2_{cal}$

Ho is accepted

⇒ Male & female births are equally probable

Q In a sample of 500 students drawn from the college students in West Bengal, it's observed that no of students reading in H.S., 1st, 2nd, 3rd year are 200, 150, 100, 50 respectively. Will it support the hypothesis that in West Bengal, the no of students reading in H.S, 1st, 2nd, 3rd year are in proportion 3:2:2:1 ? Test at 5% level of signi. Given that 5% value of x^2 for 3 & 4 dof is 7.815 and 7.416 respectively

dof $= n-1 = 4-1 = 3$

Ho : Ratio is 3:2:2:1

⇒ $x^2_{crit} = 7.815$

H_1 : Ratio isn't 3:2:2:1

	fo	Probablity	(500×P) fe	fo-fe	$(fo-fe)^2$	$\frac{(fo-fe)^2}{fe}$
HS	200	3/8	187.5	12.5	156.25	0.83
Ist	150	1/4	125	25	625	5
IInd	100	1/4	125	-25	625	5
IIIrd	50	1/8	62.5	-12.5	156.25	2.5
						13.33

$x^2_{cal} = 13.33$

$x^2_{cal} > x^2_{crit}$

H_1 accepted

Ratio isn't 3:2:2:1